

EX. NO: 4

IMPLEMENTATION OF VIGENERE CIPHER

AIM:

To implement the Vigenere Cipher substitution technique using C program.

ALGORITHM:

STEP-1: Arrange the alphabets in row and column of a 26*26 matrix.

STEP-2: Circulate the alphabets in each row to position left such that the first letter is attached to last.

STEP-3: Repeat this process for all 26 rows and construct the final key matrix. STEP-4: The keyword and the plain text is read from the user.

STEP-5: The characters in the keyword are repeated sequentially so as to match with that of the plain text.

STEP-6: Pick the first letter of the plain text and that of the keyword as the row indices and column indices respectively.

STEP-7: The junction character where these two meet forms the cipher character. STEP-8: Repeat the above steps to generate the entire cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
void encipher(); void
```

```
decipher();
```

```
int main() {    int
```

```
choice;    while (1)
```

```

    {    printf("\n1. Encrypt Text");
printf("\n2. Decrypt Text");
printf("\n3. Exit");

    printf("\n\nEnter Your Choice: ");    scanf("%d",
&choice);

    switch (choice)
    {        case 1:
encipher();        break;
        case 2:
decipher();        break;
        case 3:
exit(0);
        default:
            printf("\nInvalid Choice!\n");
        }
    }

    return 0; }

```

```

void encipher()
{    char input[50], key[50];
    int i, j;

    printf("\nEnter Plain Text: ");    scanf("%s", input);

```

```
printf("Enter Key: ");   scanf("%s", key);
```

```
printf("Encrypted Text: ");
```

```
for (i = 0, j = 0; i < strlen(input); i++, j++)
```

```
{   if (j == strlen(key))
```

```
j = 0;
```

```
    printf("%c", ((toupper(input[i]) - 'A' + toupper(key[j]) - 'A') % 26) + 'A');
```

```
}
```

```
printf("\n");
```

```
}
```

```
void decipher()
```

```
{   char input[50], key[50];
```

```
    int i, j;
```

```
printf("\nEnter Cipher Text: ");   scanf("%s", input);
```

```
printf("Enter Key: ");   scanf("%s", key);
```

```
printf("Decrypted Text: ");
```

```
for (i = 0, j = 0; i < strlen(input); i++, j++)
```

```
{   if (j == strlen(key))
```

```
j = 0;
```

```
        printf("%c", ((toupper(input[i]) - 'A' - (toupper(key[j]) - 'A') + 26) % 26) + 'A');  
    }  
  
    printf("\n");  
}
```

OUTPUT:

Output Clear

```
1. Encrypt Text  
2. Decrypt Text  
3. Exit  
  
Enter Your Choice: 1  
  
Enter Plain Text: run  
Enter Key: god  
Encrypted Text: XIQ  
  
1. Encrypt Text  
2. Decrypt Text  
3. Exit  
  
Enter Your Choice: 2  
  
Enter Cipher Text: god  
Enter Key: run  
Decrypted Text: PUQ  
  
1. Encrypt Text  
2. Decrypt Text  
3. Exit  
  
Enter Your Choice: 3  
  
=== Code Execution Successful ===
```

RESULT:

Thus the Vigenere Cipher substitution technique had been implemented successfully.